

An Analysis of Denoising diffusion probabilistic model

Denoising diffusion probabilistic model

Denoising diffusion probabilistic model -- Part 1

Architectural improvements proposed various architectural improvements. For example, they proposed log-space interpolation during backward sampling. Instead of sampling from $x_{t-1} \sim N(\mu_t(x_t, x_0), \sigma_t^2)$, they recommended sampling from $N(\mu_t(x_t, x_0), (\sigma_t^2 - \sigma_{1-v})^2)$ for a learned parameter v . In the v -prediction formalism, the noising formula $x_t = \alpha_t x_0 + (1 - \alpha_t) \epsilon$ is reparameterised by an angle ϕ_t such that $\cos \phi_t = \alpha_t$ and a "velocity" defined by $\sin \phi_t$. The network is trained to predict the velocity $\hat{v}_t$, and denoising is by $x_{t-\delta} = \cos(\delta) x_t - \sin(\delta) \hat{v}_t$. This parameterization was found to improve performance, as the model can be trained to reach total noise (i.e. $\phi_t = 90^\circ$) and then reverse it, whereas the standard parameterization never reaches total noise since $\alpha_t > 0$ is always true.

Denoising diffusion probabilistic model -- Part 2

$$\nabla_x \ln p(x|y) = \nabla_x \ln p(x) \text{ score} + \nabla_x \ln p(y|x) \text{ classifier guidance}$$
$$p(x|y) = \underbrace{\nabla_x \ln p(x)}_{\text{score}} + \underbrace{\nabla_x \ln p(y|x)}_{\text{classifier guidance}}$$

Denoising diffusion probabilistic model -- Part 3

Specifically, rectified flow seeks to match an ODE with the marginal distributions of the linear interpolation between points from distributions π_0 and π_1 . Given observations $x_0 \sim \pi_0$ and $x_1 \sim \pi_1$, the canonical linear interpolation $x_t = tx_1 + (1-t)x_0$, $t \in [0, 1]$ yields a trivial case $\dot{x}_t = x_1 - x_0$, which cannot be causally simulated without x_1 . To address this, x_t is "projected" into a

space of causally simulatable ODEs, by minimizing the least squares loss with respect to the direction $x_1 - x_0$:

Denoising diffusion probabilistic model -- Part 4

$$\nabla_{x_t} \ln q(x_t) = \frac{1}{\sigma_t^2} (-x_t + \alpha_t^{-1} E[q(x_0|x_t)])$$

Denoising diffusion probabilistic model -- Part 5

we end up with a point $x_1 \sim \pi_1$. The solution ϕ_t of the above ODE define a probability path $p_t = [\phi_t] \# \pi_0$ by the pushforward measure operator. In particular, $[\phi_1] \# \pi_0 = \pi_1$. The probability path and the velocity field also satisfy the continuity equation, in the sense of probability distribution:

Denoising diffusion probabilistic model -- Part 6

Denoising Diffusion Implicit Model (DDIM) The original DDPM method for generating images is slow, since the forward diffusion process usually takes $T \sim 1000$ to make the distribution of x_T to appear close to Gaussian. However this means the backward diffusion process also take 1000 steps. Unlike the forward diffusion process, which can skip steps as $x_t|x_0$ is Gaussian for all $t \geq 1$, the backward diffusion process does not allow skipping steps. For example, to sample $x_{t-2}|x_{t-1} \sim N(\mu_{t-2}(x_{t-1}, t-1), \Sigma_{t-2}(x_{t-1}, t-1))$ requires the model to first sample x_{t-1} . Attempting to directly sample $x_{t-2}|x_t$ would require us to marginalize out x_{t-1} , which is generally intractable. DDIM is a method to take any model trained on DDPM loss, and use it to sample with some steps skipped, sacrificing an adjustable amount of quality. If we generate the Markovian chain case in DDPM to non-Markovian case, DDIM corresponds to the case that the reverse process has variance equals to 0. In other words, the reverse process (and also the forward process) is deterministic. When using fewer sampling steps, DDIM outperforms DDPM. In detail, the DDIM sampling method is as follows. Start with the forward diffusion process $x_t = \alpha_t x_0 + \sigma_t \epsilon$. Then, during the

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backward denoising process, given x_t , $\epsilon_{\theta}(x_t, t)$ $\{\displaystyle \epsilon_{\theta}(x_t, t)\}$, the original data is estimated as $x_0' = x_t - \sigma_t \epsilon_{\theta}(x_t, t) \alpha_t^{-1}$ $\{\displaystyle x_0' = \frac{x_t - \sigma_t \epsilon_{\theta}(x_t, t)}{\alpha_t}\}$ then the backward diffusion process can jump to any step $0 \leq s < t$ $\{\displaystyle 0 \leq s < t\}$, and the next denoised sample is $x_s = \alpha_s^{-1} x_0' + \sigma_s \epsilon_{\theta}(x_s, t) + \sigma_s' \epsilon_{\theta}(x_s, t)$ $\{\displaystyle x_s = \sqrt{\alpha_s} x_0' + \sqrt{1 - \alpha_s} \epsilon_{\theta}(x_s, t)\}$ where σ_s' $\{\displaystyle \sigma_s'\}$ is an arbitrary real number within the range $[0, \sigma_s]$ $\{\displaystyle [0, \sigma_s]\}$, and $\epsilon_{\theta} \sim N(0, I)$ $\{\displaystyle \epsilon_{\theta} \sim \mathcal{N}(0, I)\}$ is a newly sampled Gaussian noise. If all $\sigma_s' = 0$ $\{\displaystyle \sigma_s' = 0\}$, then the backward process becomes deterministic, and this special case of DDIM is also called "DDIM". The original paper noted that when the process is deterministic, samples generated with only 20 steps are already very similar to ones generated with 1000 steps on the high-level. The original paper recommended defining a single "eta value" $\eta \in [0, 1]$ $\{\displaystyle \eta \in [0, 1]\}$, such that $\sigma_s' = \eta \sigma_s$ $\{\displaystyle \sigma_s' = \eta \sigma_s\}$. When $\eta = 1$ $\{\displaystyle \eta = 1\}$, this is the original DDPM. When $\eta = 0$ $\{\displaystyle \eta = 0\}$, this is the fully deterministic DDIM. For intermediate values, the process interpolates between them. By the equivalence, the DDIM algorithm also applies for score-based diffusion models.